Restricted admissibility: map and next steps

Normal experiment with $X \sim N_2(\mu, I_2)$ and $\Lambda = \{\mu_2 \le \mu_1\}$

1. The present answer to the main question

The question is not whether using X_2 is intrinsically good or bad. The right distinction is

> use X_2 only through directions that are feasible under $\mu_2 \le \mu_1$.

For $\Sigma = I_2$, the feasible nonnegative normal wedge is

$$\mathcal{F} = \{a \in \mathbf{R}^2 : 0 \le a_2 \le a_1\} = \text{cone}\{(1,0),(1,1)\}.$$

A closed convex acceptance region whose finite faces have normals in $\mathcal{F}$ is aligned with alternatives that can actually occur. The short test qualifies because its only normal is $(1,0)$. The gated-sum test

$$A_{c,b} = \{x_1 \le c,\ x_1 + x_2 \le b\}, \qquad \phi_{c,b}(x) = I\{x_1 > c \text{ or } x_1 + x_2 > b\},$$

qualifies because its finite normals are $(1,0)$ and $(1,1)$, and it genuinely uses X_2. The long test fails globally because its rectangle has the vertical face $x_2 = c_\ell$, whose normal $(0,1)$ lies outside $\mathcal{F}$.

2. What is known, what is open

Object/class	Status	Lesson
Short test $I\{X_1 > c_s\}$	Known globally admissible	Ignoring X_2 is defensible because the face normal $(1,0)$ is feasible. No class restriction is needed.
Long test $I\{\max(X_1, X_2) > c_\ell\}$	Known globally inadmissible	It uses X_2 too symmetrically. The bad feature is the $(0,1)$ face, not the mere presence of X_2.
Exact-size ordered rectangles	Known admissible among rectangles	Rectangularity forbids conditional gating. Within this restricted design class, both short and long are admissible.
Finite ordered rectangles in larger classes	Known dominated by gated sums	The rectangle's upper-right corner can be replaced by a diagonal face $x_1 + x_2 = b$, improving power on Λ_1 at the same null probability.
Gated-sum tests $I\{X_1 > c \text{ or } X_1 + X_2 > b\}$	Proof-ready/global positive class	This is the concrete admissible companion to the short test that genuinely uses X_2.
Full class "monotone + order-respecting"	Too broad	It contains the long test and its gated dominator, so closed convexity plus those two monotonicities cannot imply admissibility.
Characterization of all admissible convex tests	Open	Natural conjecture: exact-size closed convex lower tests are globally admissible iff all finite exposed normals are feasible, up to boundary/null-set caveats.

3. Relation to Andrews–Soares

Andrews–Soares write their statistic as

$$T_n(\theta) = S\{n^{1/2} m_n(\theta), \widehat{\Sigma}_n(\theta)\},$$

where, for inequality moments, Assumption 1 requires S to be nonincreasing in the inequality-moment vector, invariant to positive diagonal rescaling, nonnegative, and continuous. Under the sign convention $z_j = -m_j/\sigma_j$, their monotonicity is exactly the usual idea that larger z_j gives more evidence against the null.

The gated test is not one of their named examples S_1, S_2, S_3. But the statistic

$$S_{\text{gate},\lambda}(m, \Sigma) = \max\left\{\left[-\frac{m_1}{\sigma_1}\right]_+, \lambda\left[-\frac{m_1}{\sigma_1} - \frac{m_2}{\sigma_2}\right]_+\right\}, \qquad \lambda > 0,$$

with $\sigma_j^2 = \Sigma_{jj}$, generates the standardized gated rejection rule

$$S_{\text{gate},\lambda}(m, \Sigma) > \kappa \quad \Longleftrightarrow \quad z_1 > \kappa \quad \text{or} \quad z_1 + z_2 > \kappa/\lambda.$$

Thus, in the normal notation $x = z$, it is exactly $I\{x_1 > \kappa \text{ or } x_1 + x_2 > \kappa/\lambda\}$. For fixed λ, this statistic satisfies Assumption 1(a)–(d): it is nonincreasing in m, diagonal-scale invariant, nonnegative, and continuous. The caveat is important: it generally does *not* satisfy the Andrews–Soares positivity condition that the statistic is positive whenever any original inequality moment is violated. That is intentional here, because an X_2-only violation is not a relevant alternative under the priority restriction. So the gated statistic is Andrews–Soares-compatible at the Assumption 1 level, but it is not a standard AS statistic unless introduced as a new priority-aware S or justified by transforming/augmenting the moment vector.

4. Action plan

1. Write the feasible-normal admissibility theorem cleanly using the Lehmann–Romano ray condition. This should establish global α-admissibility for exact-size closed convex acceptance regions whose finite supporting normals lie in $\mathcal{F}$.
2. Promote the gated-sum family to the main positive example using X_2. Calibrate $b = b(c, \alpha)$ by $P_0\{X_1 \le c,\ X_1 + X_2 \le b\} = 1 - \alpha$ and choose one or two interpretable members for figures.
3. Keep the rectangle theorem, but label it correctly as design-class admissibility. Its role is to show that if the researcher insists on componentwise-threshold tests, then using X_2 via an ordered rectangle is defensible within that class.
4. Decide how much Andrews–Soares structure to claim. The safe statement is: the gated statistic satisfies Assumption 1, but may fail Assumption 3 for the unrestricted AS alternative. If Assumption 3 is needed later, either modify the moment system or explicitly say the priority model uses a different alternative.
5. Work out extensions only after the bivariate statement is stable. For $k > 2$, the feasible nonnegative normals should satisfy $a_1 \ge a_2 \ge \cdots \ge a_k \ge 0$, with prefix-sum extreme rays. For $\Sigma \ne I_2$, feasibility must be expressed through the natural-to-mean map; a natural-parameter support direction a moves the mean in direction Σa.

5. Theorems and conjectures to put in the notes

Theorem 1 (Short test: global admissibility)**.** *In the restricted experiment $X \sim N_2(\mu, I_2)$ with $\Lambda = \{\mu_2 \le \mu_1\}$, the exact-size short test $\phi_s(x) = I\{x_1 > \Phi^{-1}(1 - \alpha)\}$ is α-admissible against all level-α tests, including randomized competitors.*

Theorem 2 (Ordered rectangles: rectangular admissibility)**.** *Every exact-size ordered rectangle $A_c = \{x_1 \le c_1, x_2 \le c_2\}$ with $c_1 \le c_2$ and $\Phi(c_1)\Phi(c_2) = 1 - \alpha$ is admissible relative to all rectangular level-α competitors.*

Theorem 3 (Finite rectangles are dominated outside rectangles)**.** *For every finite exact-size ordered rectangle $A_c = \{x_1 \le c_1, x_2 \le c_2\}$, $c_1 \le c_2 < \infty$, there is a calibrated gated-sum acceptance region $A_{c_1,b} = \{x_1 \le c_1, x_1 + x_2 \le b\}$ with the same null probability whose test strictly dominates the rectangle on Λ_1.*

Theorem 4 (Feasible-normal convex sufficiency)**.** *Let A be closed and convex, let $P_0(X \in A) = 1 - \alpha$, and suppose every finite supporting normal of A lies in $\mathcal{F} = \{a : 0 \le a_2 \le a_1\}$. Then $I\{X \notin A\}$ is α-admissible against all level-α tests. In particular, every exact-size gated-sum test is globally admissible.*

Conjecture 1 (Possible characterization)**.** *Among exact-size closed convex lower acceptance regions, global admissibility is equivalent, up to null-set and boundary caveats, to having no finite exposed supporting normal outside $\mathcal{F}$.*

Proposition 1 (Andrews–Soares Assumption 1 check)**.** *For fixed $\lambda > 0$, $S_{\text{gate},\lambda}$ defined above satisfies Andrews–Soares Assumption 1(a)–(d). It is not one of S_1, S_2, S_3, and it need not satisfy their positivity condition for unrestricted moment-inequality alternatives.*